Alexander Cavoli

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Education

Rensselaer Polytechnic Institute (RPI) Troy, NY
B.S. Computer Science & Mathematics Dual Major; GPA: 3.92; Dean's Honors List *2024 – May 2028*

Experience

PresentationAssistants.com — Part-Time Developer Feb 2025 – Oct 2026
Troy, NY
- Developed classroom tools alongside Professor Jidong Xiao to support synchronized instruction and interactive student participation, helping deploy the platform across nine undergraduate computer science courses.
- Implemented a real-time collaborative code editor enabling presenters and students to edit, compile, and execute code in-browser, along with authentication, login-attempt tracking, and administrative tools for user, project, and scheduling management.
- Contributed to synchronization features for live code, presentations, and terminal sessions, supporting classroom use with 20 to approximately 300 students.

Abele Technologies — Cybersecurity Specialist Intern June 2026 – August 2026
Manchester, CT
- Supported cybersecurity and IT operations across managed client environments by developing custom security and automation tools and assisting with client-facing technical issues.
- Automated Windows disk-space remediation with a self-healing script that freed storage and generated a service ticket when usage remained above 90%. Built a safeguarded BitLocker deployment tool that enabled encryption and securely stored recovery keys in the company's password manager.

Arista Networks — Software Engineering Intern June 2024 – Aug 2024
Remote / Santa Clara, CA
- Worked under Rob Martin on `testdrive.arista.com`, updating tooling that used the `cvprac` and `rcvpapi` libraries to improve integration with Arista's CloudVision platforms.
- Refined `gitConfigletSync.py` for efficient configuration synchronization in nested lab environments, supporting automated network configuration workflows.

Publications

Presentation Assistants: A Unified Platform for Real-Time Synchronization and Learner-Controlled Navigation ACM SIGCSE 2027
- **First author** of a Peer-reviewed paper accepted to the ACM SIGCSE Technical Symposium 2027.
- Evaluated a real-time classroom platform through a two-year deployment across nine undergraduate CS courses, analyzing student engagement, participation, and interaction with synchronized instructional content.

Projects

4-time hackathon winner; selected projects highlighted below.

Cornell Hackathon Winner — AI Auto Sort Rover | *Python, Arduino, CLIP* Oct 2026
- Won Best Hardware Hack and placed 4th overall among 120+ projects at Cornell's BigRedHacks hackathon with an AI-powered mobile waste sorting rover.
- Integrated locally running CLIP computer vision with Raspberry Pi-controlled servo trash can lids to identify waste and automatically open the corresponding trash or recycling bin.
- Built a web dashboard using PostgreSQL to track disposal history, recycling rewards, and detection analytics; integrated wireless rover control through keyboard, voice commands, and hand gestures.

University of Delaware Hackathon Winner — TrailMate.Tech | *Node.js, JavaScript, CSS, Snowflake* Mar 2026
- Built a campus walking companion designed to help students meet new people and make walking more social, contributing across front-end UI, backend logic, and database integration.
- Integrated AR pathing with real-time guidance, building names, and distance updates; implemented live walk tracking, interactive maps, and AI-driven trend detection using Snowflake.

Technical Skills

Languages: Python, C++, C, C#, JavaScript, SQL, PHP, HTML/CSS
Tools & Technologies: Git, Linux, Node.js, MySQL, WebSockets, WebAssembly, REST APIs, Snowflake